

Course: Computer Graphics Lab

Lab Setup

CodeBlocks + OpenGL (FreeGLUT) Setup Tutorial

Pre-Requirements

Download these files first:

1. CodeBlocks with MinGW

- File: codeblocks-20.03mingw-setup.exe

2. Notepad++

- File: npp.8.9.3.Installer.x64.exe

3. FreeGLUT Package

- File: freeglut-mingw-master
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Installation & Setup Process

Step 1 — Copy FreeGLUT DLL Files

Open the freeglut-mingw-master folder.

Go to:

bin

Copy all files and paste them into:

C:\Windows

Step 2 — Copy Header Files

Go to:

Include\GL

Copy all files.

Paste them into:

C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\include\GL

Step 3 — Copy Library Files

Go to:

lib

Copy all files.

Paste them into:

C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\lib



Modify CodeBlocks GLUT Template

Step 4 — Edit `glut.cbp`

Open this folder:

C:\Program Files\CodeBlocks\share\CodeBlocks\templates

Find:

`glut.cbp`

Open it using Notepad++

Replace Text

Press:

Ctrl + H

Find:

glud32

Replace with:

freeglut

Save the file.

Step 5 — Edit `wizard.script`

Go to:

C:\Program Files\CodeBlocks\share\CodeBlocks\templates\wizard\glut

Find:

wizard.script

Open it using Notepad++

Replace Text

Press:

Ctrl + H

Find:

glud32

Replace with:

freeglut

Save the file.



Configure CodeBlocks

Step 6 — Open CodeBlocks

Launch Code::Blocks

Create a New GLUT Project

Follow These Steps:

File → New → Project → GLUT Project

Click:

Go

Then complete the project setup.

Configure Compiler Path

Go to:

Settings → Compiler → Toolchain Executables

Set the compiler path correctly if needed.

Usually:

C:\Program Files\CodeBlocks\MinGW

Test OpenGL Program

Use this simple program to check whether OpenGL works correctly:

```
#include <GL/glut.h>
```

```
void display()
```

```
{
```

```
    glClear(GL_COLOR_BUFFER_BIT);
```

```
    glBegin(GL_TRIANGLES);
```

```
    glColor3f(1,0,0);
```

```
    glVertex2f(-0.5,-0.5);
```

```
    glColor3f(0,1,0);
```

```
    glVertex2f(0.5,-0.5);
```

```
    glColor3f(0,0,1);
    glVertex2f(0,0.5);

    glEnd();

    glFlush();
}

int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutCreateWindow("OpenGL Test");

    glutDisplayFunc(display);

    glutMainLoop();

    return 0;
}
```

Expected Output

If everything is configured correctly:

- ☒ A window will open
 - ☒ A colorful triangle will appear
 - ☒ OpenGL + FreeGLUT setup is complete
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Common Errors & Fixes

 **undefined reference to glut**

Fix:

Check whether:

- freeglut library files are copied correctly
 - glut32 was replaced with freeglut
-

 **GL/glut.h not found**

Fix:

Header files were not copied properly into:

include/GL

✗ Window opens and closes instantly

Fix:

Use:

```
glutMainLoop();
```

inside main().

🏁 Final Result

After completing all steps successfully:

- ☒ CodeBlocks ready
- ☒ MinGW configured
- ☒ FreeGLUT installed
- ☒ OpenGL projects can run perfectly 🚀